IMPERIAL

Department of Computing
70015: Mathematics for Machine Learning
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Practice Problems 04: Automatic Differentiation

Question 38 (Product rule). (I) Consider the function $f(a, b) = a \cdot b$, where $a = a(x), b = b(x)$, i.e. unspecified functions of x.

- Show that by following forward mode autodiff, you effectively calculate the product rule.
- Show that if $a(x) = x, b(x) = x$, which means that the overall function $f(x) = x^2$, the gradient that is computed will be $2x$.

Question 39 (Log Normal Density). (II) Draw the computational graph for the log density, $L(y; \mu, \sigma^2) = \log\left(\mathcal{N}(y; \mu, \sigma^2)\right)$. *(Hint: it is useful to simplify your above answer as much as possible.)* Write out the forward mode updates for the value tangent given that we are interested in computing $\dfrac{\partial L}{\partial \mu}$.

Question 40 (Chain rule). (I) Consider the function $f(x) = g(h(x))$, where $h(x)$ and $g(x)$ are unspecified functions of x.

(i) Write down the forward mode autodiff updates for $h(x)$, $g(h(x))$, and their derivatives.

(ii) Show that the derivative computed by forward mode autodiff corresponds to the chain rule:

$$\frac{\mathrm{d}f}{\mathrm{d}x} = g'(h(x)) \cdot h'(x).$$

(iii) If $h(x) = x^2$ and $g(z) = \sqrt{z}$, find the gradient of $f(x) = \sqrt{x^2}$ and verify the result manually.

Question 41 (Backward mode autodiff). (II) Consider the function $f(x, y) = (x + y)^2 + \sin(xy)$.

(i) Draw the computational graph for $f(x, y)$, clearly labeling all intermediate nodes.

(ii) Perform backward mode autodiff to compute the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

(iii) Verify your results by manually calculating $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Question 42 (Backward mode autodiff with matrix operations). (II) Consider the function $f(\mathbf{A}, \mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$, where:

- $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a matrix,
- $\mathbf{x} \in \mathbb{R}^n$ is a vector,
- $\mathbf{b} \in \mathbb{R}^m$ is a fixed vector, and
- $\|\cdot\|^2$ denotes the squared Euclidean norm.

(i) Draw the computational graph for $f(\mathbf{A}, \mathbf{x})$, labeling all intermediate variables and their shapes.

(ii) Compute the gradient $\nabla_{\mathbf{x}} f$ using backward mode autodiff, and discuss the computational complexity of each step in terms of m and n.

(iii) Compute the gradient $\nabla_{\mathbf{A}} f$ using backward mode autodiff, and discuss the computational complexity of each step in terms of m and n.

(iv) Discuss how the shapes of the gradients align with the shapes of $\mathbf{A}$ and $\mathbf{x}$.

Question 43 (Multivariate Autodiff)**.** (IV) This question is extremely hard and is not representative of questions you would be asked on an exam. Completion of this question represents mastery of the topic. This is a rather big question that should test your understanding of all material in the first three lectures. Consider the overall function $f(\boldsymbol{\ell}, \mathrm{X})$ consisting of the parts:

$$f = \mathbf{y}^\intercal (\mathrm{K}_1 + \mathrm{K}_2)^{-1} \mathbf{y}\,, \tag{4.16}$$

$$\mathrm{K}_a = \exp(\Lambda_a)\,, \tag{4.17}$$

$$\Lambda_a = -\frac{\mathrm{D}_a}{2\ell_a^2}\,, \tag{4.18}$$

$$\mathrm{D}_a = (\mathrm{X}[:, \mathsf{None}, a] - \mathrm{X}[\mathsf{None}, :, a])^2\,, \tag{4.19}$$

where we use `numpy` broadcasting notation in the final equation.

a. Given $\boldsymbol{\ell} \in \mathbb{R}^2$ and $\mathrm{X} \in \mathbb{R}^{N \times 2}$, find the shape of all intermediate computations.

b. Draw the computational graph for $f(\boldsymbol{\ell}, \mathrm{X})$.

c. For forward and reverse mode differentiation, state which intermediate derivatives are computed at each step, and their computational and memory costs.